

LOWERING THE FIELD-SIZE THRESHOLD IN THE POLYNOMIAL TWIN-PRIME THEOREM

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ABSTRACT. Let p be an odd prime and let q be a power of p . We combine Sawin's pointwise estimate for the von Mangoldt function in squarefree arithmetic progressions with the Mobius–prime correlation estimate of Sawin and Shusterman. In the von Mangoldt–Mobius convolution for twin prime polynomials, the former estimate treats divisor degrees $k \leq 5d/7$, while the latter treats $k > 5d/7$. This gives the Hardy–Littlewood asymptotic over $\mathbb{F}_q[T]$ under the condition

$$q > \frac{1156}{9} e^2 p^2,$$

and hence under the simpler condition $q > 950p^2$. This lowers the sufficient condition $q > 685090p^2$ in the theorem of Sawin and Shusterman. As a consequence, every nonzero polynomial is a sum of two irreducible elements of arbitrarily large equal degree. No new geometric estimate is proved here; the point is the compatibility of two published estimates in complementary ranges of the convolution.

1. INTRODUCTION

Let $\mathcal{M}_d$ denote the set of monic polynomials of degree d in $\mathbb{F}_q[T]$, and let Λ and μ be the polynomial von Mangoldt and Mobius functions. Sawin and Shusterman proved a quantitative twin-prime theorem over $\mathbb{F}_q[T]$ when $q = p^r$, p is odd, and

$$(1.1) \quad q > 685090p^2.$$

More precisely, their Theorem 6.3 gives a Hardy–Littlewood asymptotic with a power-saving error term [4, Theorem 6.3]. Their proof expands one von Mangoldt function by

$$(1.2) \quad \Lambda = -1 * (\mu \deg)$$

and splits the resulting divisor variable into two ranges. The small-divisor range is handled by a level of distribution for primes, whereas the large-divisor range uses a Mobius–prime correlation estimate that breaks the parity barrier.

Sawin subsequently proved a stronger pointwise estimate for the von Mangoldt function in progressions to squarefree moduli [3, Theorem 1.2 and Proposition 8.16]. The squarefree restriction causes no loss in (1.2), since the divisor variable is weighted by μ . Replacing only the small-divisor input

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in the proof of [4, Theorem 6.3] permits a much later splitting point. The large-divisor parity-breaking argument is left unchanged.

For a nonzero monic $h \in \mathbb{F}_q[T]$, put

$$(1.3) \quad \mathfrak{S}_q(h) = \prod_{P|h} (1 - |P|^{-1})^{-1} \prod_{P \nmid h} \left(1 - (|P| - 1)^{-2}\right), \quad |P| = q^{\deg P},$$

where P runs over monic irreducible polynomials. Our parameterized form of the observation is as follows.

Theorem 1.1 (Hybrid criterion). *Let p be an odd prime, let $q = p^r$, and fix $0 < \omega < 1$. Put*

$$(1.4) \quad \theta = \frac{1}{2 - \omega}.$$

Suppose that

$$(1.5) \quad q > 4^{\frac{2-\omega}{1-\omega}} \left(\frac{2-\omega}{1-\omega} - \frac{1}{2} \right)^2 e^2,$$

$$(1.6) \quad q > \left(2pe \left(\frac{4}{\omega} - 1 \right) \right)^2.$$

Then, for every fixed nonzero monic $h \in \mathbb{F}_q[T]$, there is a $\lambda = \lambda(q, \omega) > 0$ such that

$$(1.7) \quad \sum_{f \in \mathcal{M}_d} \Lambda(f) \Lambda(f + h) = \mathfrak{S}_q(h) q^d + O_{q,h,\omega} \left(q^{(1-\lambda)d} \right)$$

as $d \rightarrow \infty$.

The clean numerical consequence is the following.

Corollary 1.2 (The $950p^2$ threshold). *The conclusion of [theorem 1.1](#) holds if*

$$(1.8) \quad q > \frac{1156}{9} e^2 p^2.$$

In particular, it holds whenever

$$(1.9) \quad \boxed{q > 950p^2}.$$

Proof. Take $\omega = 3/5$, so that $\theta = 5/7$. The right-hand side in (1.5) is

$$4^{7/2} \left(\frac{7}{2} - \frac{1}{2} \right)^2 e^2 = 1152e^2,$$

whereas the right-hand side in (1.6) is

$$\left(2pe \left(\frac{20}{3} - 1 \right) \right)^2 = \frac{1156}{9} e^2 p^2.$$

Since $p \geq 3$, the latter is larger than the former. Finally,

$$\frac{1156}{9} e^2 = 949.083205 \dots < 950.$$

□

The argument is a new combination of published estimates, not a new cohomological construction. To the best of our knowledge, the combination and the numerical threshold in [corollary 1.2](#) have not previously appeared in print. Bagshaw later obtained strong distribution estimates for arbitrary composite moduli [1]; he explicitly notes that Sawin's theorem gives the more relaxed field-size condition when the modulus is squarefree [1, §2.3]. A disabled passage in the public arXiv source of [1], absent from both the compiled preprint and the journal article, records that Bagshaw's estimate could instead lower (1.1) to $181157p^2$.¹ Thus the general idea of replacing the small-divisor input was present in that source, while the sharper squarefree substitution and the constant (1.8) appear not to have been recorded.

2. THE TWO PUBLISHED INPUTS

We state only the special cases needed below. Throughout, $\varphi(M)$ is the number of units in $\mathbb{F}_q[T]/(M)$.

Proposition 2.1 (Sawin). *Fix $0 < \theta < 1$ and suppose that*

$$(2.1) \quad q > 4^{\frac{1}{1-\theta}} \left(\frac{1}{1-\theta} - \frac{1}{2} \right)^2 e^2.$$

Then there is a $\delta = \delta(q, \theta) > 0$ such that, whenever $m \leq \theta n$, $M \in \mathcal{M}_m$ is squarefree, and $a \in (\mathbb{F}_q[T]/(M))^\times$, one has

$$(2.2) \quad \sum_{\substack{f \in \mathcal{M}_n \\ f \equiv a \pmod{M}}} \Lambda(f) = \frac{q^n}{\varphi(M)} + O_{q,\theta} \left((q^{n-m})^{1-\delta} \right).$$

One may take

$$(2.3) \quad \delta = \frac{1}{2} - \log_q \left(\frac{2^{\theta/(1-\theta)} e(1+\theta)}{1-\theta} \right) > 0.$$

Proof. This is [3, Theorem 1.2 and Proposition 8.16]. □

The second input is the specialization of the Mobius–prime correlation estimate used by Sawin and Shusterman in the large-divisor part of their twin-prime proof.

Proposition 2.2 (Sawin–Shusterman). *Fix $0 < \alpha < 1$. Suppose that there is a $0 < \beta < 1/2$ for which*

$$(2.4) \quad q > \left(2pe \frac{3+\alpha}{1-\alpha} \right)^{\frac{2}{1-2\beta}}.$$

Let h be fixed and monic. Uniformly for $N \in \mathcal{M}_m$ with $(N, h) = 1$ and integers $m \leq \alpha k$ with $\deg h < m + k$, one has

$$(2.5) \quad \sum_{M \in \mathcal{M}_k} \mu(M) \Lambda(h + MN) \ll_{q,h,\alpha,\beta} q^{k(1-\frac{\beta(1-\alpha)}{2p})}.$$

¹See the [arXiv source archive](#), file `Files/comments.tex`; the passage is enclosed in `\iffalse...\fi`.

Proof. This follows from the $n = 1$ specialization of [4, Corollary 6.1], with the auxiliary degree parameters taken sufficiently large. We include a normalization detail that is implicit in the application [4, (6.22)–(6.24)]. Choose $j \in \{1, 2\}$ such that

$$\deg h \neq m + j,$$

and put $C = T^j$. For all sufficiently large k , translation by C is a bijection of $\mathcal{M}_k$, and

$$\sum_{M \in \mathcal{M}_k} \mu(M) \Lambda(h + MN) = \sum_{g \in \mathcal{M}_k} \mu(C + g) \Lambda((h + CN) + gN).$$

Both fixed terms C and $h + CN$ are monic. The pairs $(C, 1)$ and $(h + CN, N)$ are coprime and distinct because $(h, N) = 1$ and $h \neq 0$. Corollary 6.1 now applies with variable degree k , $n = 1$, and prime-form modulus degree m . The strict inequality (2.4) absorbs the harmless finite choices of its auxiliary parameters. $\square$

We will also use the following truncation of the singular series.

Proposition 2.3 (Sawin–Shusterman). *For fixed q , a nonzero monic h , and $K \geq 1$,*

$$(2.6) \quad \sum_{k=1}^K k \sum_{\substack{M \in \mathcal{M}_k \\ (M, h)=1}} \frac{\mu(M)}{\varphi(M)} = -\mathfrak{S}_q(h) + (q-1)^{-K+o(K+\deg h)}.$$

Proof. This is [4, Proposition 6.2]. It follows by differentiating the Euler product

$$\prod_{P \nmid h} \left(1 - \frac{u^{\deg P}}{|P| - 1} \right) = (1 - u)G_h(u),$$

evaluating $u \frac{d}{du}((1 - u)G_h(u))$ at $u = 1$, and estimating the tail on circles of radius less than $q - 1$. $\square$

3. THE HYBRID CONVOLUTION

Fix a nonzero monic h . Since h is fixed, we assume throughout that $d > \deg h$. Define

$$C_d(h) = \sum_{f \in \mathcal{M}_d} \Lambda(f) \Lambda(f + h).$$

The convolution identity (1.2) gives

$$(3.1) \quad C_d(h) = - \sum_{k=1}^d k \sum_{M \in \mathcal{M}_k} \mu(M) \sum_{N \in \mathcal{M}_{d-k}} \Lambda(h + MN).$$

Let

$$(3.2) \quad K = \lfloor \theta d \rfloor, \quad \theta = \frac{1}{2 - \omega}.$$

We estimate separately the ranges $k \leq K$ and $k > K$.

3.1. The small-divisor range. If $\mu(M) \neq 0$, then M is squarefree. For $(M, h) = 1$, the map

$$N \mapsto h + MN$$

is a bijection from $\mathcal{M}_{d-k}$ to the monic degree- d polynomials in the residue class h modulo M . Since $k \leq K \leq \theta d$, [proposition 2.1](#) gives

$$(3.3) \quad \sum_{N \in \mathcal{M}_{d-k}} \Lambda(h + MN) = \frac{q^d}{\varphi(M)} + O_{q,\theta} \left(q^{(d-k)(1-\delta)} \right).$$

The total progression error in the small-divisor range is at most

$$(3.4) \quad \begin{aligned} \sum_{k \leq K} k \sum_{M \in \mathcal{M}_k} q^{(d-k)(1-\delta)} &\leq d \sum_{k \leq K} q^k q^{(d-k)(1-\delta)} \\ &\ll d^2 q^{d-\delta(d-K)} \\ &\ll q^{d-\delta(1-\theta)d+o(d)}. \end{aligned}$$

This summation is the only additional estimate needed to pass from Sawin's pointwise progression theorem to the full convolution.

For completeness, the omitted moduli with $(M, h) \neq 1$ are negligible. Indeed, if $P \mid (M, h)$ and $\Lambda(h + MN) \neq 0$, then $h + MN = P^j$ for some $j \geq 1$. For a fixed $P \mid h$ and d , there is at most one such prime power of degree d , and the possible factorizations $MN = P^j - h$ are bounded by the divisor function of a degree- d polynomial. The elementary bound $\tau(A) \leq 2^{\deg A}$ therefore shows that the full non-coprime contribution to [\(3.1\)](#) is $O_h(d^2 2^d)$. This is a power saving because $q \geq 3$.

Using [proposition 2.3](#), the main terms from [\(3.3\)](#) sum to

$$(3.5) \quad \begin{aligned} -q^d \sum_{k \leq K} k \sum_{\substack{M \in \mathcal{M}_k \\ (M, h)=1}} \frac{\mu(M)}{\varphi(M)} &= \mathfrak{S}_q(h) q^d + O \left(q^d (q-1)^{-K+o(K+\deg h)} \right) \\ &= \mathfrak{S}_q(h) q^d + O_{q,h} \left(q^{(1-c_0)d} \right) \end{aligned}$$

for some $c_0 = c_0(q, \theta) > 0$. Together, [\(3.4\)](#) and [\(3.5\)](#) evaluate the small-divisor range with a power-saving error.

3.2. The large-divisor range. If $k > K$, then

$$(3.6) \quad \frac{d-k}{k} < 1 - \omega.$$

Set $\alpha = 1 - \omega$. The strict hypothesis [\(1.6\)](#) allows us to choose $\beta > 0$ sufficiently small that

$$q > \left(2pe \left(\frac{4}{\omega} - 1 \right) \right)^{\frac{2}{1-2\beta}} = \left(2pe \frac{3+\alpha}{1-\alpha} \right)^{\frac{2}{1-2\beta}}.$$

For every $N \in \mathcal{M}_{d-k}$ with $(N, h) = 1$, [proposition 2.2](#) and [\(3.6\)](#) therefore give

$$(3.7) \quad \sum_{M \in \mathcal{M}_k} \mu(M) \Lambda(h + MN) \ll q^{(1-\eta)k}, \quad \eta = \frac{\beta\omega}{2p} > 0.$$

The terms with $(N, h) \neq 1$ are negligible by the same prime-power argument used in the small-divisor range. Namely, if $P \mid (N, h)$ and $\Lambda(h + MN) \neq 0$,

then $h + MN = P^j$; summing the factorizations of $P^j - h$ over the finitely many $P \mid h$ gives $O_h(d^2 2^d)$. Summing over N and k gives

$$\begin{aligned}
 \sum_{k>K} k \sum_{N \in \mathcal{M}_{d-k}} \left| \sum_{M \in \mathcal{M}_k} \mu(M) \Lambda(h + MN) \right| &\ll d \sum_{k>K} q^{d-k} q^{(1-\eta)k} \\
 &\ll d^2 q^{d-\eta K} \\
 &\ll q^{d-\eta \theta d + o(d)}.
 \end{aligned}
 \tag{3.8}$$

This is again a power saving. Combining (3.4), (3.5), and (3.8) proves [theorem 1.1](#).

4. PRIME PAIRS AND A GOLDBACH COROLLARY

Let $\mathcal{I}_d$ be the set of monic irreducible polynomials of degree d .

Corollary 4.1 (Unweighted twin primes). *Under the hypotheses of [theorem 1.1](#), there is a $\lambda' > 0$ such that*

$$(4.1) \quad \#\{f \in \mathcal{I}_d : f + h \in \mathcal{I}_d\} = \mathfrak{S}_q(h) \frac{q^d}{d^2} + O_{q,h,\omega} \left(q^{(1-\lambda')d} \right).$$

Proof. A degree- d polynomial counted by Λ but not irreducible is a proper prime power. There are $O_q(q^{d/2})$ such polynomials, and the same bound applies after translation by h . Their total contribution to (1.7) is $O_q(d^2 q^{d/2})$. Dividing the remaining weighted count by d^2 proves the result after slightly decreasing λ . $\square$

Since q is odd, the Euler product (1.3) converges to a positive number. Thus [corollary 4.1](#) supplies twin irreducibles for every sufficiently large d . It also has the following ring-theoretic additive consequence.

Corollary 4.2 (Ring-theoretic $1 + 1$). *Assume (1.8). Every nonzero $F \in \mathbb{F}_q[T]$ is a sum of two irreducible elements of arbitrarily large equal degree.*

Proof. Let $c = \text{lc}(F)$ and let $h = c^{-1}F$, which is monic. By [corollary 4.1](#), for every sufficiently large d there is an $f \in \mathcal{I}_d$ such that $f + h \in \mathcal{I}_d$. Hence

$$F = c(f + h) + (-cf).$$

Both summands are irreducible elements of $\mathbb{F}_q[T]$, being unit multiples of monic irreducibles, and both have degree d . $\square$

Remark 4.3. The representation in [corollary 4.2](#) uses cancellation of the leading terms. It is therefore not the degree-preserving Goldbach normalization of Bender and Pollack, in which a monic polynomial of degree n is written as a sum of monic irreducibles of degrees $n - 1$ and n [2]. Sawin and Shusterman state that their method also establishes a function-field Goldbach analogue [4, Remark 1.2], but they do not record that normalization and its error term as a separate theorem. We make no claim about the corresponding numerical threshold here.

5. NUMERICAL COMPARISON

The choice $\omega = 3/5$ is made for a clean uniform constant, not because it is exactly optimal. The following table compares the least powers of p satisfying the displayed sufficient conditions (1.1) and (1.8).

Characteristic p	Sawin–Shusterman condition	corollary 1.2 condition
3	$\mathbb{F}_{3^{15}}$	$\mathbb{F}_{3^9}$
5	$\mathbb{F}_{5^{11}}$	$\mathbb{F}_{5^7}$
7	$\mathbb{F}_{7^9}$	$\mathbb{F}_{7^6}$
11	$\mathbb{F}_{11^8}$	$\mathbb{F}_{11^5}$

These are only the least powers satisfying the stated sufficient inequalities; they are not asserted to be the smallest fields over which the twin-prime asymptotic is true. One may lower the numerical constant slightly by balancing (1.5) and (1.6). For example, at $p = 3$ the balancing value is

$$\omega = 0.6002068419\dots,$$

which gives a threshold $8534.8241\dots$, or $948.3138\dots p^2$. The rational choice $\omega = 3/5$ loses very little and yields the transparent bound $950p^2$.

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